

Solutions to STAT 306 Midterm Exam

Problem 1. Marking scheme: 70% for correct procedures, and 30% for correct answers.

(a) For a male teacher with three years of service, we have: $x_2 = 1$, $x_1 = 3$. For a female teacher with three years of experience, we have $x_2 = 0$, $x_1 = 3$. Thus, the difference in mean salaries is given by

$$(45 + 3 * 3 + 4 * 1 + 1 * 3 * 1) - (45 + 3 * 3 + 4 * 0 + 1 * 3 * 0) = 7 \text{ (in thousands of dollars, or \$7,000)}$$

(an answer of -7 or $-7,000$ are also acceptable.)

(b) For a female teacher with three years of experience, we have $x_2 = 0$, $x_1 = 3$. Thus, the predicted/estimated salary for the female teacher with three years of service is

$$45 + 3 * 3 + 4 * 0 + 1 * 3 * 0 = 54.$$

Thus, the residual of this observation is

$$y_i - \hat{y}_i = 60 - 54 = 6$$

(in thousands of dollars, or \$6,000). (an answer of -6 or $-6,000$ are also acceptable).

(c) We can test $H_0 : \beta_3 = 0$ versus $H_1 : \beta_3 \neq 0$. The test statistic is

$$z = \frac{\hat{\beta}_3}{SE(\hat{\beta}_3)} = \frac{1}{0.8} = 1.25$$

Since $1.25 < 1.96$ (or 2), we fail to reject H_0 , i.e., the interaction term x_1x_2 is not significant. Thus, the relationship between salary and years of service **does not** depend on gender.

(d) (2 pts each) (i) In (c), since the interaction term is not significant, we may remove it from model (1). Thus, a simpler model is

$$y = \beta_0 + \beta_1x_1 + \beta_2x_2 + e.$$

(ii) If $\hat{\beta}_3 = 4$ (0.8), the test statistic is

$$z = \frac{\hat{\beta}_3}{SE(\hat{\beta}_3)} = \frac{4}{0.8} = 5 > 1.96 \text{ (or 2),}$$

the interaction is significant, so the relationship between salary and years of service depends on gender. In this case, it's recommended to study the relationship for male and female separately.

Problem 2.

(a) (2 pts for (i), 3 pts for (ii)) (i) Interpretation for $\hat{\beta}_1$ in Model (2): One hour increase per week in exercise time is associated 9.48 pounds weight loss over a year, adjusted for sugar consumption (or hold sugar consumption as fixed, or incorporating sugar consumption information).

(ii) In Model (2), the relationship between weight loss and exercise time is adjusted for sugar consumption, while in Model (1) sugar consumption is ignored.

(b) (2 pts for (i) and 3 pts for (ii)) (i) One possible reason is that exercise time and sugar consumption may be highly correlated (so the effect of exercise time on weight loss is greatly affected by whether sugar consumption is taken into account or not).

(ii) If the values of $\hat{\beta}_1$ are very close in the two models, exercise time and sugar consumption may be weakly correlated (or relatively independent) so that adding one in the model does not affect the effect of the other one.

(c) (2 pts for (i) and 3 pts for (ii)) (i) If the relationship in Model (1) is not linear, the residual plot should show a clear curved pattern, such as a quadratic pattern.

(ii) If the variance of the random error increases with weight loss, the residuals should *increase* with fitted values of weight loss. A log-transformation of the response $\log(y)$ may possibly fix this problem.

Problem 3.

(a) Compared with low sugar consumption, medium sugar consumption is associated with 1.51 weight **gain**, adjusted for exercise (or holding exercise time fixed).

Compared with low sugar consumption, high sugar consumption is associated with 2.01 weight **gain**, adjusted for exercise (or holding exercise time fixed).

(b) For the categorical variable sugar consumption x_2 , compared to the reference category (low sugar consumption), medium sugar consumption seems to be significantly associated with weight gain, but high sugar consumption seems not significant (compared with critical values -1.96 or -2 , with significance level 0.05). The overall significance x_2 in Model (3) is *unclear* based on the information given. Further hypothesis testing based on other methods is needed.

(c) No. Model (3) is **not** equivalent to ANOVA, since Model (3) incorporates exercise information while ANOVA ignores this information.

(d) (2 pts for each question) (1) With y-axis as weight loss and x-axis as exercise time, (i) should show **three parallel upward lines**, and (ii) should show that the three lines in (i) are **not parallel** (may be crossed).

(2). For (i), the model is

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + e.$$

For (ii), the model is

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1 x_2 + e.$$